

Beamer⁺

Interactive Presentations

Chandra Gummaluru

`chandra-gummaluru.github.io`

Beamer⁺ is a lightweight web app that enhances static slides with embedded media, real-time drawing tools, intuitive navigation, and a suite of powerful presentation features.

using Beamer⁺

the steps to create a Beamer⁺ presentation

(1) **create**

create slides using any standard tool (including L^AT_EX)

(2) **configure**

write `config` files to specify the placement and behaviour of any embedded media

(3) **package**

combine the slides, embedded media, and `config` files into a package

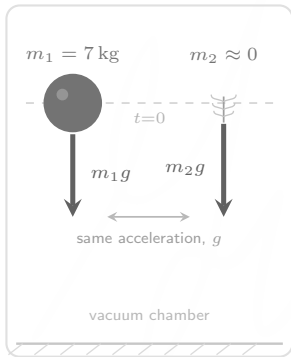
(4) **present**

open the package with Beamer⁺, which will display the slides with all embedded media integrated seamlessly

videos/audio

supported formats include .mp4, .mov, .mpeg, .mp3, .wav

example usage: demonstrating Newton's 2nd law in a physics course

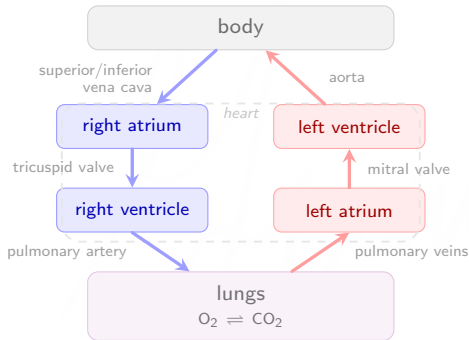


gravitational acceleration is
independent of mass

models

supported formats include .glpb

example usage: explaining blood flow through the heart in an anatomy course



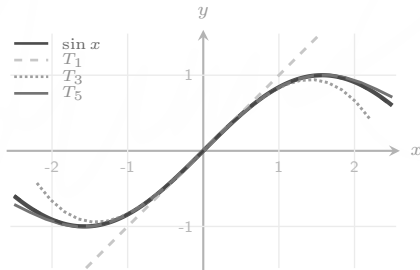
blood flow through the heart
(deoxygenated blood / oxygenated blood)

supported formats include `.html`

example usage: explaining Taylor series in a calculus class

Any smooth function f can be approximated around $x_0 = 0$ as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n$$



expansion of $f(x) = \sin(x)$ around $x_0 = 0$



widget library

a collection of commonly used tools

examples include: calculator, timer, and camera)

(calculator widget)

(timer widget)

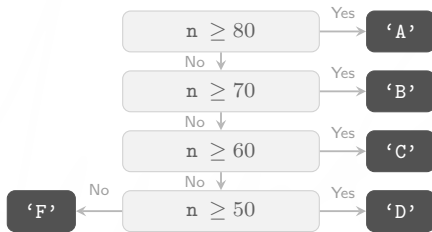
(camera widget)

use in computer science

a collection of commonly used tools
(e.g., calculator, timer, webcam, etc.)

(survey widget)

A flowchart for an algorithm to convert a numerical grade, n , to a letter grade is shown below:



numerical to letter grade conversion algorithm













learn more

visit the link below to learn more

`chandra-gummaluru.github.io/post/beamer-plus`